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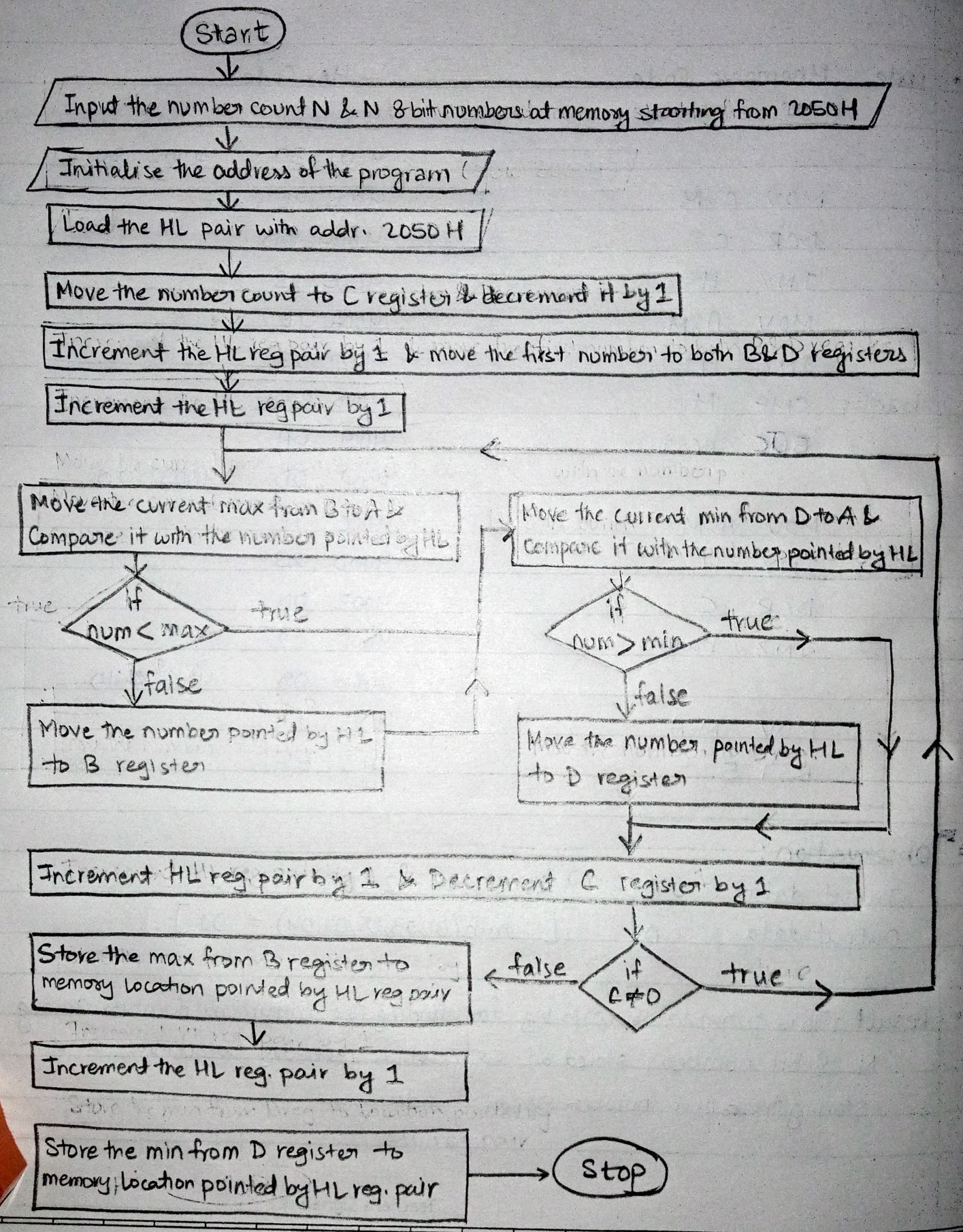
- Aim: Write a program in assembly Language of 8085 to find the max & the min numbers among N 8-bit numbers stored at consecutive memory locations.

- Registers used : HL pair : to point memory address
  C reg : to store number count
  Accumulator : to compare the numbers
  B reg : to store current maximum
  D reg : to store current minimum

- Algorithm : (i) start and initiate the address of the program.
(ii) Load the HL reg pair with address 2050H, where number count is stored.
(iii) move the number count to C register, which we later use as counter.
(iv) decrement C by 1, because for N numbers we have to make (N-1) comparisons.
(v) increment HL reg. pair to the next address, where first number is stored.
(vi) move the first num to both B & D registers, these are initial max & min.
(vii) increment HL reg pair to the next address, to start comparisons.
(viii) move current max from B to A, & compare it with the number pointed by HL reg pair
(ix) if the pointed number is smaller, jump to step (xi)
(x) else, update the current max.
(xi) move current min from D to A, & compare it with the number pointed by HL reg. pair
(xii) if the pointed number is greater, jump to step (xiv)
(xiii) else, update the current min.
(xiv) increment HL reg. pair to the next address, & decrement C reg by 1.
(xv) if C is not zero, jump to step (viii)
(xvi) store the max from B reg, to the memory location pointed by HL reg. pair

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- Flow Chart


Start
Input the number count N & N 8 bit numbers at memory starting from 2050H
Initialise the address of the program
Load the HL pair with addr. 2050H
Move the number count to C register & decremant H by 1
Increment the HL reg pair by 1 & move the first number to both B & D registers
Increment the HL reg pair by 1
Move the current max from B to A & Compare it with the number pointed by HL
if num < max
true
false
Move the number pointed by HL to B register
Move the current min from D to A & Compare it with the number pointed by HL
if num > min
true
false
Move the number pointed by HL to D register
Increment HL reg. pair by 1 & Decrement C register by 1
if C≠0
false
true
Store the max from B register to memory location pointed by HL reg pair
Increment the HL reg. pair by 1
Store the min from D register to memory location pointed by HL reg. pair
Stop

- Observation Table

Memory Addr	Before	After	Decimal equiv.
2050	05	05	5 (number count)
2051	02	02	2
2052	04	04	4
2053	01	01	1
2054	05	05	5
2055	03	03	3
2056	00	05 (max)	∵ max(2,4,1,5,3)=5
2057	00	01 (min)	∵ min(2,4,1,5,3)= 1

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(xvii) Increment HL reg pair to the next address.

(xviii) store the min from D reg, to the memory location pointed by HL reg pair

(xix) stop

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Code	Mnemonic Code	Hex Code	
	LXI H,2050H	3000 : 21	
		3001 : 50	3002 : 20
	MOV C,M	3003 : 4E	
	DCR C	3004 : 0D	
	INX H	3005 : 23	
	MOV B,M	3006 : 46	
	MOV D,M	3007 : 56	
	INX H	3008 : 23	
back :	MOV A,B	3009 : 78	
	CMP M	300A : BE	
	JNC next1	300B : D2	
		300C : 0F	300D : 30
	MOV B,M	300E : 46	
next1 :	MOV A,D	300F : 7A	
	CMP M	3010 : BE	
	JC next2	3011 : DA	
		3012 : 15	3013 : 30
	MOV D,M	3014 : 56	
next2 :	INX H	3015 : 23	
	DCR C	3016 : 0D	
	JNZ back	3017 : C2	
		3018 : 09	3019 : 30

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MOV M,B	301A : 70
INX H	301B : 23
MOV M,D	301C : 72
RST 5	301D : EF

- Observation :

  Input data → (05), 02,04,01,05,03

  Output data → max = 05

  min = 01

- Result : This program is working good for finding the maximum & minimum numbers among N 8-bit numbers stored at consecutive memory locations, and storing them at next free addresses.

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